Effects of a low-salt diet on flow-mediated dilatation in humans.

BACKGROUND: The effect of salt reduction on vascular function, assessed by brachial artery flow-mediated dilatation (FMD), is unknown. OBJECTIVE: Our aim was to compare the effects of a low-salt (LS; 50 mmol Na/d) diet with those of a usual-salt (US; 150 mmol Na/d) diet on FMD. DESIGN: This was a randomized crossover design in which 29 overweight and obese normotensive men and women followed an LS diet and a US diet for 2 wk. Both diets had similar potassium and saturated fat contents and were designed to ensure weight stability. After each intervention, FMD, pulse wave velocity, augmentation index, and blood pressure were measured. RESULTS: FMD was significantly greater (P = 0.001) with the LS diet (4.89 +/- 2.42%) than with the US diet (3.37 +/- 2.10%), systolic blood pressure was significantly (P = 0.02) lower with the LS diet (112 +/- 11 mm Hg) than with the US diet (117 +/- 13 mm Hg), and 24-h sodium excretion was significantly lower (P = 0.0001) with the LS diet (64.1 +/- 41.3 mmol) than with the US diet (156.3 +/- 56.7 mmol). There was no correlation between change in FMD and change in 24-h sodium excretion or change in blood pressure. No significant changes in augmentation index or pulse wave velocity were observed. CONCLUSIONS: Salt reduction improves endothelium-dependant vasodilation in normotensive subjects independently of the changes in measured resting clinic blood pressure. These findings suggest additional cardioprotective effects of salt reduction beyond blood pressure reduction. The trial is registered with the Australian and New Zealand Clinical Trials Registry (unique identifier: ANZCTR12607000381482; http://www.anzctr.org.au/trial_view.aspx?ID=82159).